

Diagonalizing a Matrix

Previously, $Ax = \lambda x$.

$$\text{Also, } A^2x = \lambda^2x$$

$$A^{-1}x = \frac{1}{\lambda}x.$$

If we have $x_1, x_2, x_3, \dots, x_n$ for $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$
then let S be the eigenvector vector where,

$$S = \begin{bmatrix} | & | & | & \dots & | \\ x_1 & x_2 & x_3 & \dots & x_n \\ | & | & | & \dots & | \end{bmatrix}$$

$$\text{Now, } AS = \begin{bmatrix} | & | & | & \dots & | \\ \lambda_1 x_1 & \lambda_2 x_2 & \lambda_3 x_3 & \dots & \lambda_n x_n \\ | & | & | & \dots & | \end{bmatrix}$$

$$= \begin{bmatrix} | & | & | & \dots & | \\ x_1 & x_2 & x_3 & \dots & x_n \\ | & | & | & \dots & | \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Λ (capital λ)

$$\text{Here, } AS = S\Lambda$$

$$\Rightarrow S^{-1}AS = \Lambda$$

$$\text{Also, } A = S\Lambda S^{-1} \quad \text{⊗}$$

$$\hookrightarrow A^2 = (S\Lambda S^{-1})(S\Lambda S^{-1})$$

$$\rightarrow A^2 = S\Lambda^2 S^{-1}$$

$$\rightarrow A^3 = (S\Lambda^2 S^{-1})(S\Lambda S^{-1})$$

$$\rightarrow A^3 = S\Lambda^3 S^{-1}$$

o

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$$\rightarrow A^k = S \Lambda^k S^{-1} \quad (\text{X})$$

Some key points.

i) Any matrix that has no repeated eigenvalues can be diagonalized.

$\rightarrow$ If eigenvalues are different numbers, then eigenvectors will be independent. Thus eigenvector matrix X will be invertible.

ii) We can multiply eigenvectors by any non-zero constants

$$\rightarrow A(cx) = \lambda(cx)$$

iii) The eigenvalues in Λ come in the same order as the eigenvectors in X .

$\rightarrow \lambda_1, \lambda_2$ for a_1, a_2 will look like:

$$\begin{bmatrix} 1 & 1 \\ \lambda_1 a_1 & \lambda_2 a_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \quad \text{not} \quad \begin{bmatrix} \lambda_2 & 0 \\ 0 & \lambda_1 \end{bmatrix}$$

iv) Matrices that have too few eigenvectors cannot be diagonalized.

- If there is only one eigenvalue, it cannot be diagonalized.

(X) Diagonalizability requires n linearly independent eigenvectors for an $n \times n$ matrix. S has to be invertible (no repeated eigenvalues)

v) $\rightarrow$

- **Invertibility** depends on the eigenvalues. A matrix is invertible if and only if it **does not** have **zero** as an eigenvalue ($\lambda \neq 0$).
- **Diagonalizability** depends on the number of independent eigenvectors. Even if a matrix is invertible, it **might not be diagonalizable** if it lacks a full set of linearly independent eigenvectors.

vi) $A^k \Rightarrow$ zero matrix, $|\lambda| < 1$.

Similar Matrices : Same Eigenvalues

A and B are similar if $B = M^{-1}AM$

Similar matrices have the same eigenvalues.

If $Bx = M^{-1}AMx = \lambda x$ then $A(Mx) = \lambda(Mx)$

For example $\begin{bmatrix} 2 & 3 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -4 & 6 \end{bmatrix}$

↳ They are all similar matrices. They are a family of similar matrices with eigenvalues 2 and 4.

⊗ But what does similar matrices actually represent?

↳ They represent the same linear transformation under different bases.

Suppose a vector v is transformed by A : $v' = Av$

Now represent v in a basis defined by M : $u = Mv$.

Then if we apply M on v' ,

$$\left. \begin{aligned} Mu' &= MAv \\ \Rightarrow Mu' &= MAM^{-1}u \\ \Rightarrow u' &= (MAM^{-1})u \end{aligned} \right\} \begin{array}{l} \text{Since, } u = Mv \\ \Rightarrow v = M^{-1}u \\ \text{Assume } u' = Mv' \end{array}$$

$$\hookrightarrow B = M^{-1}AM$$

(Not MAM^{-1} because it
applies transformation in the wrong order,
I got no convincing explanation for this yet)

⊗ Properties of similar matrices:

- i) Same eigenvalues
- ii) Same characteristics polynomial $\Rightarrow \det(A - \lambda I) = \det(B - \lambda I)$
- iii) Same determinant and trace. $\Rightarrow \det(A) = \det(B)$, $\text{tr}(A) = \text{tr}(B)$
- iv) Different eigenvectors $\Rightarrow$ Eigenvectors can change under similarity transformation
- v) Same Diagonalizability $\Rightarrow$ If one is diagonalizable, the other is too.

⊗ Now for some matrices, there could be repeated eigenvalues.

For example: $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $\begin{bmatrix} 0 & 2 \\ -2 & 4 \end{bmatrix} \rightarrow \text{similar}$
 $\lambda = 2, 2$ $\lambda = 2, 2$ $\rightarrow$ has one missing eigenvector

Jordan Form

$\hookrightarrow$ A special similar matrix (J)

$\hookrightarrow$ If J is diagonalizable, then its Jordan form is simply a diagonal matrix.

$\hookrightarrow$ Jordan form mainly deals with non-diagonalizable matrix.

↳ Jordan Form mainly deals with non-diagonalizable matrix.

For example:

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \rightsquigarrow \text{diagonalizable} \\ \lambda = 4, 4 \rightsquigarrow \text{two eigenvectors}$$

$$B = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix} \rightsquigarrow \text{not diagonalizable} \\ \rightsquigarrow \text{one eigenvector} \\ \lambda = 4, 4$$

↳ Jordan Form

Even though A and B have same eigenvalues,
they are not similar because not both are diagonalizable.

So for $\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix} \rightarrow$

$$\begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 17 & 4 \end{bmatrix}, \begin{bmatrix} a & x \\ y & 8-a \end{bmatrix} \rightsquigarrow \text{they all are similar} \\ \rightsquigarrow \text{none of them are diagonalizable.}$$

⊛ The Jordan form of a matrix A consists

of Jordan blocks along the diagonal.

If $A = \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$ then Jordan form, J is $\begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix} J_5(2)$

If $A = \begin{bmatrix} 5 & 4 & 2 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix}$ then Jordan form, J is $\begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} J_5(3)$

(We won't need much of Jordan form,
just have a surface level idea)

⊛ If A is $m \times n$ and B is $n \times m$, AB and BA have the same non zero eigenvalues.

Proof:

⊙ Here $AB \rightarrow m \times m$

$BA \rightarrow n \times n$

⊙ Let us construct, $D = \begin{bmatrix} I & A \\ 0 & I \end{bmatrix}$ (this is invertible and its determinant is 1)

$$E = \begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix}$$

Our goal is to show that this matrix is similar

$\sim \dots \sim P^{-1}EP$ (not done)

Can you do this

to another matrix F . Here, $F = D^{-1} E D$ (not dead)

① If $D = \begin{bmatrix} I & A \\ 0 & I \end{bmatrix}$, then $D^{-1} = \begin{bmatrix} I & -A \\ 0 & I \end{bmatrix}$

$$\begin{aligned} \text{Then, } D^{-1} E &= \begin{bmatrix} I & -A \\ 0 & I \end{bmatrix} \begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix} \\ &= \begin{bmatrix} AB - AB & 0 \\ B & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ B & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \hookrightarrow D^{-1} E D &= \begin{bmatrix} 0 & 0 \\ B & 0 \end{bmatrix} \begin{bmatrix} I & A \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix} \end{aligned}$$

This equation $F = D^{-1} E D$ says F is similar to E

So, they have the same eigenvalues.

$\begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix} \rightsquigarrow$ has the m eigenvalues
of AB plus n zeroes

$\begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix} \rightsquigarrow$ has the n eigenvalues
of BA plus m zeroes

So, AB and BA has the same eigenvalues,

the only difference is the number of zeroes which is: $|m - n|$

Fibonacci Numbers

0, 1, 1, 2, 3, 5, 8, 13 comes from $F_{k+2} = F_{k+1} + F_k$

The sequence here is obvious and you all understand it.

What if you were asked to find F_{100} or maybe F_{1000} ? The slow approach is to just keep adding them, but that is not definitely what we want.

Let's see what we can do:

i) Start with a vector u_0 .

Let's say, $u_1 = Au_0$

$$u_2 = Au_1 = A \cdot Au_0 = A^2 u_0$$

$$u_3 = Au_2 = A \cdot A^2 u_0 = A^3 u_0$$

$\vdots$

$$u_{k+1} = Au_k = A \cdot A^k u_0 = A^{k+1} u_0$$

ii) Write u as a combination of eigenvectors:

$$u_0 = c_1 x_1 + c_2 x_2 + c_3 x_3 + \dots + c_n x_n = Sc$$

$$\hookrightarrow Au_0 = c_1 \lambda_1 x_1 + c_2 \lambda_2 x_2 + c_3 \lambda_3 x_3 + \dots + c_n \lambda_n x_n$$

$$\hookrightarrow A^{100} u_0 = c_1 \lambda_1^{100} x_1 + c_2 \lambda_2^{100} x_2 + c_3 \lambda_3^{100} x_3 + \dots + c_n \lambda_n^{100} x_n$$

$$= A^{100} Sc \quad \leadsto \text{A generalized form}$$

iii) According to Fibonacci rules:

$$F_{k+2} = F_{k+1} + F_k \quad \text{--- (i)}$$

$$F_{k+1} = F_{k+1} + 0 \quad \text{--- (ii) } \leadsto \text{Dummy Equation}$$

Let $u_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$ (Here we are putting two consecutive Fibonacci numbers in a vector)

So, from (i) and (ii), we can construct $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

$$u_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} u_k = A u_k \leadsto u_{100} = A^{100} u_0$$

$$u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, u_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \dots u_{100} = \begin{bmatrix} F_{101} \\ F_{100} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\hookrightarrow |A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 1$$

$$\hookrightarrow \lambda^2 - \lambda - 1 = 0 \Rightarrow \lambda = \frac{1 \pm \sqrt{5}}{2} \quad (\text{Golden ratio! } \textcircled{00})$$

$$\hookrightarrow \lambda_1 = \frac{1}{2}(1 + \sqrt{5}), \lambda_2 = \frac{1}{2}(1 - \sqrt{5})$$

$$\approx 1.618 \dots \quad \approx -0.618 \dots$$

So, A is diagonalizable for two distinct eigenvalues.

Now, $A - \lambda I = \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix}$

For λ_1 , $\begin{bmatrix} 1-\lambda_1 & 1 \\ 1 & -\lambda_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\hookrightarrow (1-\lambda_1)x_1 + x_2 = 0 \quad \text{--- (i)}$
 $x_1 - \lambda_1 x_2 = 0 \quad \text{--- (ii)}$

Setting $x_2 = 1$ in (ii) $\Rightarrow x_1 = \lambda_1$

Also for λ_2 , $x_1 = \lambda_2$

$\therefore$ The eigenvectors are $\begin{bmatrix} \lambda_1 \\ 1 \\ v_1 \end{bmatrix}$ and $\begin{bmatrix} \lambda_2 \\ 1 \\ v_2 \end{bmatrix}$

Now, going back to u_0 ,

$\rightarrow$ express u_0 as $u_0 = \alpha v_1 + \beta v_2$

$\rightarrow$ solving for α and β :

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \alpha \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}$$

$\hookrightarrow \begin{cases} \alpha \lambda_1 + \beta \lambda_2 = 1 & \text{--- (i)} \\ \alpha + \beta = 0 & \Rightarrow \alpha = -\beta & \text{--- (ii)} \\ & \Rightarrow \beta = -\alpha \end{cases}$

$\begin{cases} \alpha \lambda_1 - \alpha \lambda_2 = 1 \\ \Rightarrow \alpha = \frac{1}{\lambda_1 - \lambda_2} \end{cases}$

$\Rightarrow \alpha = \frac{1}{\sqrt{5}}$

$\begin{cases} -\beta \lambda_1 + \beta \lambda_2 = 1 \\ \Rightarrow \beta = \frac{1}{\lambda_2 - \lambda_1} \end{cases}$

$\Rightarrow \beta = -\frac{1}{\sqrt{5}}$

$\lambda_1 = \frac{1}{2}(1+\sqrt{5})$

$\lambda_2 = \frac{1}{2}(1-\sqrt{5})$

$\therefore \lambda_1 - \lambda_2 = \sqrt{5}$

$\lambda_2 - \lambda_1 = -\sqrt{5}$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2} \left(\begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix} - \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix} \right) \quad \text{or} \quad u_0 = \frac{v_1 - v_2}{\lambda_1 - \lambda_2} \\ = \alpha \lambda_1 v_1 - \beta \lambda_2 v_2$$

So, for $u_{100} = A^{100} u_0$

$$\Rightarrow u_{100} = \alpha \lambda_1^{100} v_1 - \beta \lambda_2^{100} v_2$$

$$\Rightarrow u_{100} = \frac{(\lambda_1)^{100} v_1 - (\lambda_2)^{100} v_2}{\lambda_1 - \lambda_2}$$

$$\hookrightarrow \text{For } F_{100} = \frac{(\lambda_1)^{100} - (\lambda_2)^{100}}{\lambda_1 - \lambda_2} \\ = \frac{\left\{ \frac{1}{2}(1 + \sqrt{5}) \right\}^{100}}{\sqrt{5}} \quad \text{Here, } (\lambda_2)^{100} \approx 0$$

$$= 354224848179263111169$$

(obviously you don't need to write the value)

So, for any $F_n = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n$.

$$\otimes \lim_{n \rightarrow \infty} \frac{F_{n+1}}{F_n} = \frac{1 + \sqrt{5}}{2} \approx 1.618 \leadsto \text{Golden Ratio}$$

Matrix Powers A^k

We have expressed u_0 as a combination of eigenvectors:

$$u_0 = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$$

$$\textcircled{i} \quad C = X^{-1} u_0 \quad \left| \begin{array}{l} \text{Here,} \\ C \rightarrow \text{co-efficient} \\ X \rightarrow \text{matrix of eigenvectors} \end{array} \right.$$

$\textcircled{ii}$ Eigenvectors of A satisfy $Ax_i = \lambda_i x_i$, applying A^k :

$$A^k(x_i) = (\lambda_i)^k x_i$$

$$\text{So, } u_k = A^k u_0 = c_1 (\lambda_1)^k x_1 + c_2 (\lambda_2)^k x_2 + \dots + c_n (\lambda_n)^k x_n$$

$$\therefore u_{k+1} = A^k u_k \quad \left(\text{Here, } A^k = X \Lambda^k X^{-1} \right)$$

Non-Diagonalizable Matrices

Definition of Eigenvalues and Eigenvectors

1. Eigenvectors (Geometric View):

- These are solutions to $Ax = \lambda x$.
- The number of independent eigenvectors is crucial for diagonalization.

2. Eigenvalues (Algebraic View):

- Found by solving $\det(A - \lambda I) = 0$.
- If an eigenvalue appears multiple times, it has a multiplicity (AM).

Multiplicity of an Eigenvalue

- If λ is an eigenvalue, it can appear multiple times.
- We classify its multiplicity in two ways:
 1. Geometric Multiplicity (GM): Number of independent eigenvectors.
 2. Algebraic Multiplicity (AM): Number of times λ appears as a root of $\det(A - \lambda I) = 0$.
- Always:

$$GM \leq AM$$

(Geometric multiplicity is always less than or equal to algebraic multiplicity.)

Example of a Repeated Eigenvalue

- Suppose A has an eigenvalue $\lambda = 4, 4, 4$.
- Then $\lambda = 4$ has $AM = 3$ (since it appears 3 times).
- GM can be 1, 2, or 3 depending on the number of independent eigenvectors.

Key Example of a Non-diagonalizable Matrix

- The matrix:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

has:

- Eigenvalue $\lambda = 0$ with $AM = 2$.
- But only one eigenvector, so $GM = 1$.
- Since $GM < AM$, the matrix is not diagonalizable.

⊗ A matrix is diagonalizable only if $GM = AM$ for all eigenvalues.

⊗ If $GM < AM$, the matrix cannot be diagonalized.

Questions and Proofs of Eigenvalues and Eigenvectors.

Q1: Does all square matrices have eigenvectors and eigenvalues?

Answer: All square matrices have eigenvalues. It depends on their characteristics polynomial whether the eigenvalues are real or not. Also there will be at least one eigenvector corresponding to one eigenvalues.

Q2: Only symmetric square matrices would have a set of real, orthonormal eigenvectors, why?

Ans: Let A be a symmetric matrix. So, $A^T = A$.

Now, $Av = \lambda v$ $\hookrightarrow$ ①

$\hookrightarrow$ transposing from both sides. $v^T A^T = v^T \lambda$

$$\Rightarrow v^T A = v^T \lambda' \quad \text{--- (ii)}$$

Now,

$$v^T A v - v^T A = v^T \lambda v - v^T \lambda'$$

$$\Rightarrow v^T \lambda' - v^T A = v^T \lambda v - v^T \lambda' v$$

$$\Rightarrow v^T v (\lambda - \lambda') = 0$$

Since v is non zero, therefore it must be greater than 0.

So, $\lambda - \lambda' = 0 \Rightarrow \lambda = \lambda' \therefore$ They are real eigenvalues.

Assume v_1 and v_2 are eigenvectors of A which is a symmetric matrix.

$$\hookrightarrow Av_1 = \lambda_1 v_1, Av_2 = \lambda_2 v_2$$

$$\hookrightarrow v_2^T Av_1 = v_2^T \lambda_1 v_1 \Rightarrow v_2^T A v_1 = \lambda_1 (v_2^T v_1)$$

$$\hookrightarrow v_2^T A^T v_1 = \lambda_1 (v_2^T v_1) \quad [\because A^T = A]$$

$$\hookrightarrow (A v_2)^T v_1 = \lambda_1 (v_2^T v_1) \quad [\because (AB)^T = B^T A^T]$$

$$\hookrightarrow (\lambda_2 v_2)^T v_1 = \lambda_1 (v_2^T v_1)$$

$$\hookrightarrow \lambda_2 (v_2^T v_1) = \lambda_1 (v_2^T v_1)$$

$$\hookrightarrow (\lambda_2 - \lambda_1) (v_2^T v_1) = 0$$

$$\hookrightarrow v_2^T v_1 = 0 \quad [\lambda_2 - \lambda_1 \neq 0] \leadsto \text{So, } v_2^T v_1 = 0$$

$$\therefore v_2 \perp v_1$$

Q3: Prove that, $\det(A) = \lambda_1 \lambda_2 \dots \lambda_n$

Proof: $A = P \Lambda P^{-1}$

Assume that λ contains complex eigenvalues. In that case, complex eigenvalues will appear in conjugate pair. If $\lambda = 1+i$, $\lambda' = 1-i$. λ can be real, if and only if $\lambda = \lambda'$

Proof:

$$A = S \Lambda S^{-1}$$

$$\Rightarrow \det(A) = \det(S \Lambda S^{-1})$$

$$\Rightarrow \det(A) = \det(S) \cdot \det(\Lambda) \cdot \det(S^{-1}) \quad [\because \det(AB) = \det(A) \cdot \det(B)]$$

$$\Rightarrow \det(A) = \det(S) \cdot \det(\Lambda) \cdot \frac{1}{\det(S)} \quad [\because \det(S^{-1}) = \frac{1}{\det(S)}]$$

$$\Rightarrow \det(A) = \det(\Lambda)$$

$$[\because \Lambda = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}]$$

$$\Rightarrow \det(A) = \lambda_1 \lambda_2 \dots \lambda_n$$

$$\therefore A = \lambda_1 \lambda_2 \dots \lambda_n$$

Q4: Prove that, $\text{trace}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$.

Proof:

$$A = S \Lambda S^{-1}$$

$$\Rightarrow \text{tr}(A) = \text{tr}(S \Lambda S^{-1})$$

According to the cyclic property for trace:

$$\text{tr}(XYZ) = \text{tr}(YZX) = \text{tr}(ZXY)$$

$$\therefore \operatorname{tr}(S\Lambda S^{-1}) = \operatorname{tr}(\Lambda S^{-1}S) = \operatorname{tr}(S^{-1}S\Lambda)$$

$$\therefore \operatorname{tr}(\Lambda) = \operatorname{tr}(\Lambda S^{-1}S)$$

$$\Rightarrow \operatorname{tr}(A) = \operatorname{tr}(\Lambda)$$

$$\Rightarrow \operatorname{tr}(\Lambda) = \lambda_1 + \lambda_2 + \dots + \lambda_n \quad \left[\because \Lambda = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} \right]$$

$$\therefore \operatorname{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n \quad (\text{proved})$$

Q5: $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$, What is A^k if $k = 2025$?

Solⁿ:

$$\det(A - \lambda I) = 0$$

$$\hookrightarrow \begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda) \{ (5-\lambda)(1-\lambda) - 1 \} \\ - \{ (1-\lambda) - 3 \} + 3 \{ 1 - 3(5-\lambda) \} = 0$$

$$\Rightarrow (1-\lambda) (5 - 5\lambda - \lambda + \lambda^2 - 1) \\ - (-2-\lambda) + 3(1 - 15 + 3\lambda) = 0$$

$$\Rightarrow (1-\lambda)(\lambda^2-6\lambda+4) + 2+\lambda+9\lambda-42=0$$

$$\Rightarrow \lambda^2-6\lambda+4-\lambda^3+6\lambda^2-4\lambda+10\lambda-40=0$$

$$\Rightarrow \lambda^3-7\lambda^2+30\lambda-40=0$$

$$\Rightarrow \lambda = -2, 6, 3.$$

Now, $\lambda_1 = -2$

$$(A - \lambda_1 I) x_1 = 0$$

$$\hookrightarrow \begin{bmatrix} 3 & 1 & 3 \\ 1 & 7 & 1 \\ 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$\hookrightarrow$ refers to infinite solution,
so taking a suitable basis:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Now, $\lambda_2 = 6$

$$(A - \lambda_2 I)u = 0$$

$$\hookrightarrow \begin{bmatrix} -5 & 1 & 3 \\ 1 & -1 & 1 \\ 3 & 1 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$\hookrightarrow$ same case

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Now, $\lambda_3 = 3$

$$\hookrightarrow (A - \lambda_3 I)u = 0$$

$$\hookrightarrow \begin{bmatrix} -2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$\hookrightarrow$ same case

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$A = S \Lambda S^{-1}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -2 & 0 & 0 \\ 9 & 6 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -1 \\ -1 & 1 & 1 \end{bmatrix}^{-1}$$